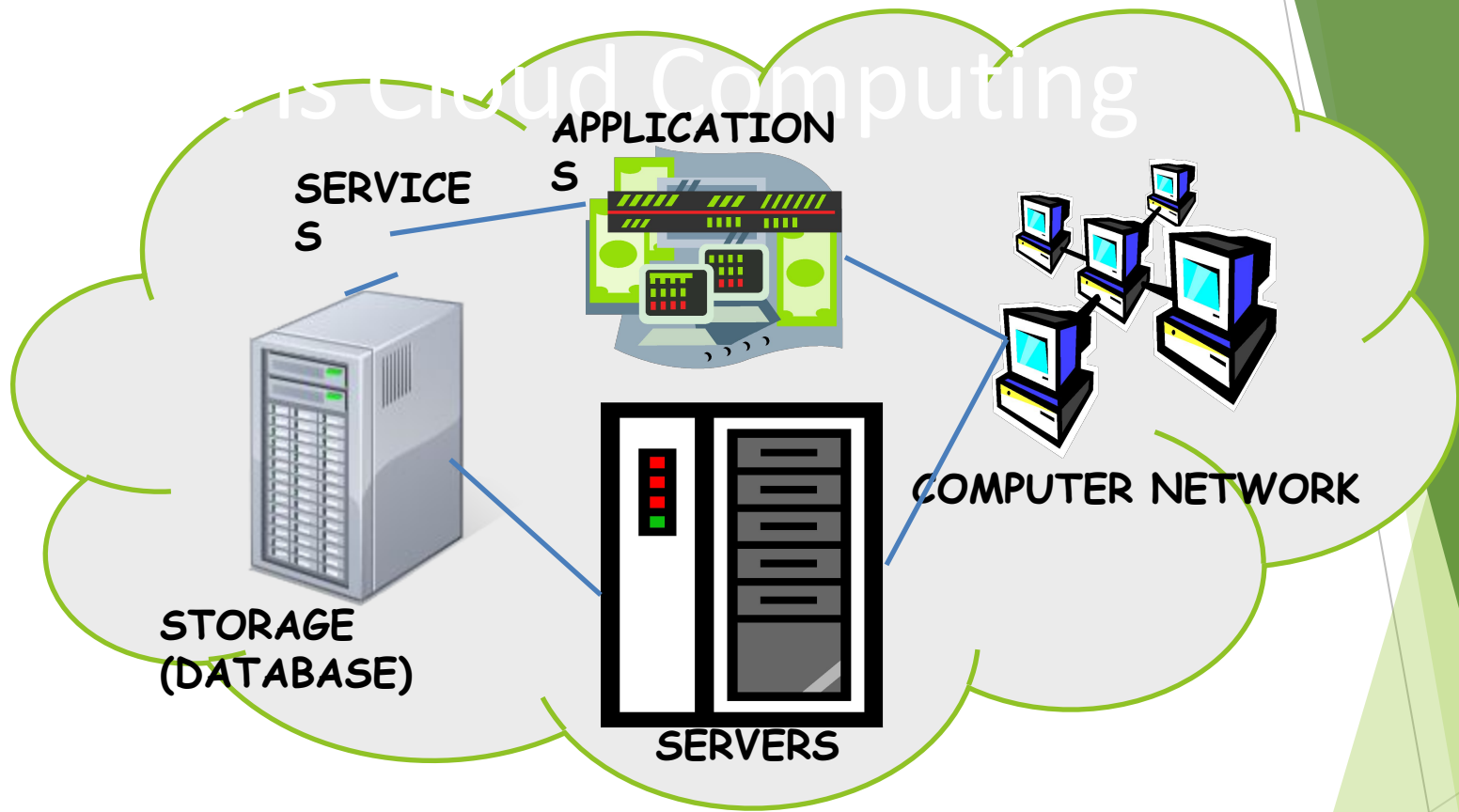


Unit-I Basics of Cloud Computing

- **Cloud Computing** is a general term used to describe a new class of network based computing that takes place over the Internet,
 - basically a step on from Utility Computing
 - a collection/group of integrated and networked hardware, software and Internet infrastructure (called a platform).
 - Using the Internet for communication and transport provides hardware, software and networking services to clients
- These platforms hide the complexity and details of the underlying infrastructure from users and applications by providing very simple graphical interface or API (Applications Programming Interface).

- In addition, the platform provides on demand services, that are always on, anywhere, anytime and any place.
- Pay for use and as needed, elastic
 - scale up and down in capacity and functionalities
- The hardware and software services are available to
 - general public, enterprises, corporations and businesses markets



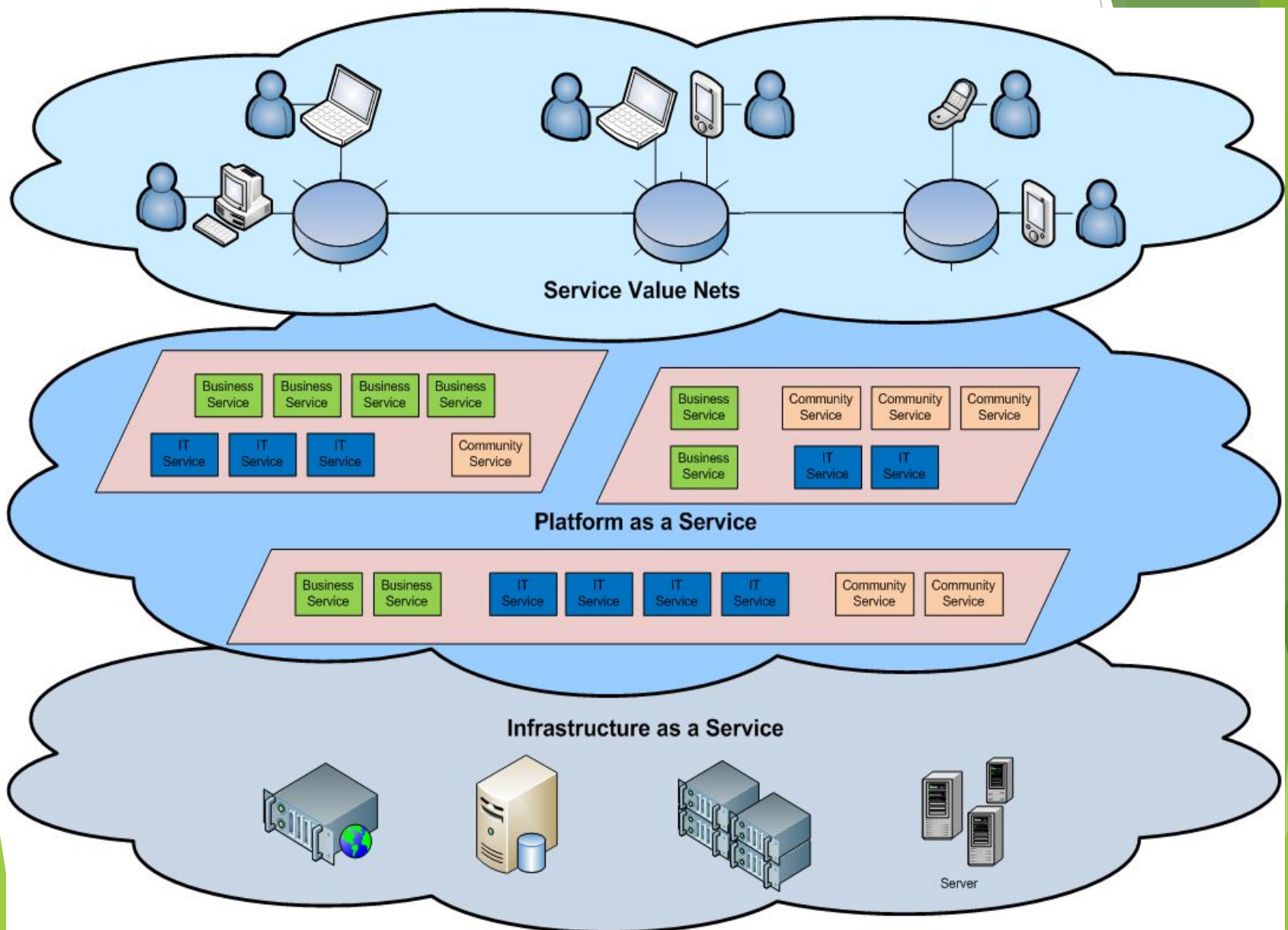
- Shared pool of configurable computing resources
- On-demand network access

Cloud Definition By NIST

The U.S. National Institute of Standards and Technology (NIST) defines cloud computing as:

- Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

- Rapid elasticity
 - You can go from 5 servers to 50 or from 50 servers to 5
- Measured service
 - You pay for what you use
- On-demand self-service
 - You get elasticity automatically
- Ubiquitous network access
 - You can access the cloud from anywhere
- Location-independent resource pooling
 - You work with virtual machines that could be hosted anywhere



CLOUD COMPONENTS

- **3 components**
 - Clients
 - Datacenter
 - Distributed servers

CLOUD COMPONENTS

- Clients

- Mobile

- SmartPhones, Tablets, Service Hubs

- Thin

- no internal hard drives, lets servers do all work, displays info

- Thick

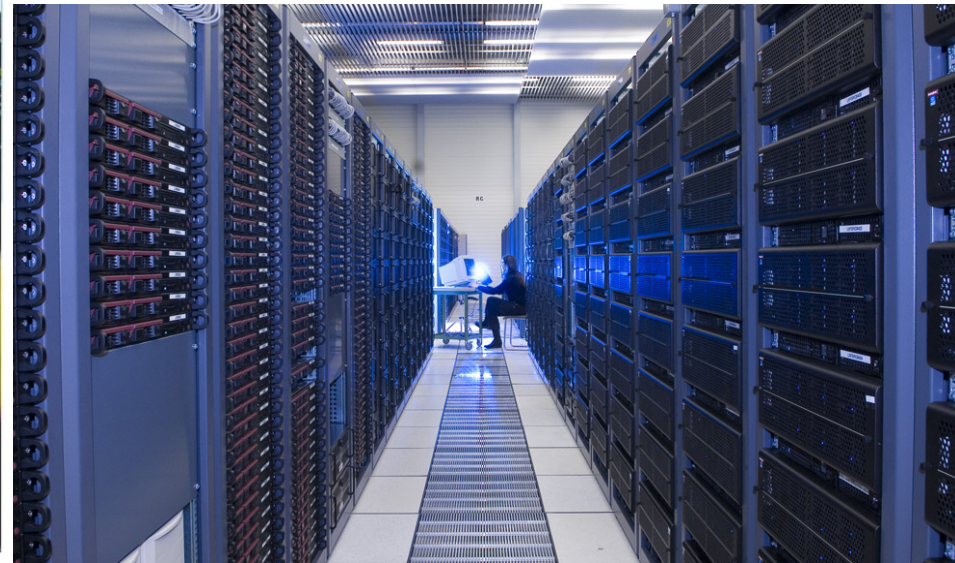
- Laptops, desktop computers

- Which is the best?

- Thin - lower costs, security, power consumption, easy to replace, less noise

DATA CENTER

- Data Center - facility used to house computer systems and associated components



DISTRIBUTED SERVERS

- Servers host the resources needed by cloud users
 - Compute nodes
 - Provides CPU, Memory, Scratch Storage, and Networking resources through virtualized interfaces.
 - Hosts guest operating systems (Virtual Machines) using one or more VM hypervisors
 - Resource interface depends on the type of cloud (horizontal/vertical cloud)

• Storage nodes

- Compute nodes only provide temporary storage space for users/applications
- Storage nodes provide long term data storage solutions
- Can be mapped to specific processes running on compute nodes, users, interface applications, etc.

• Administrative nodes

- Provides “hidden” back-end services such as resource load balancing, administrative/resource databases, security/firewalls, cloud macro management

TYPES OF CLOUDS

- **Public Cloud**

- Marketed based on

- Resources offered, availability, security, price

- **Local/Private Cloud**

- Cloud architectures tailored to an organization's needs.

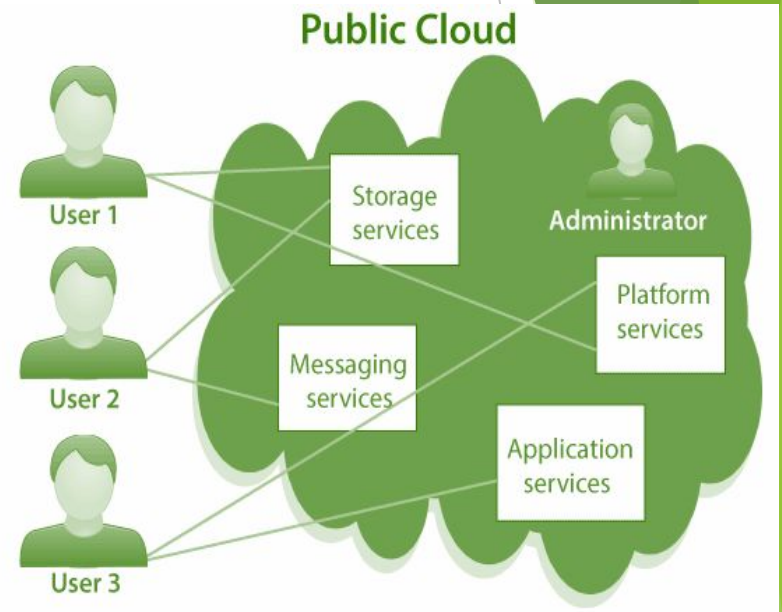
- **Hybrid Cloud**

- Combination of public and local cloud resources.

Cloud Deployment Models

Public Cloud

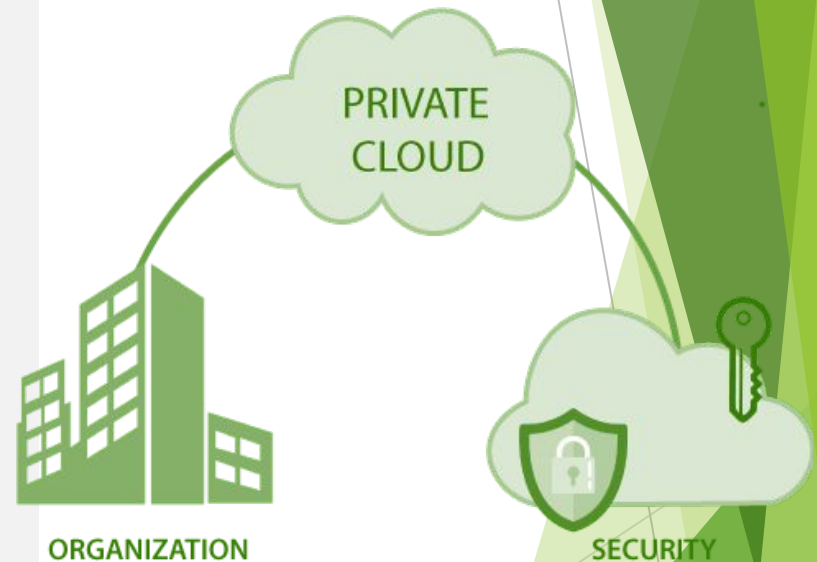
- Services are provided over a public network and available to anyone who wants to use them.
- It is a cost-effective option for businesses and individuals looking for scalability and flexibility.
- Public cloud providers, such as AWS, Azure, and GCP, offer a wide range of services accessible to the general public.



Cloud Deployment Models

Private Cloud

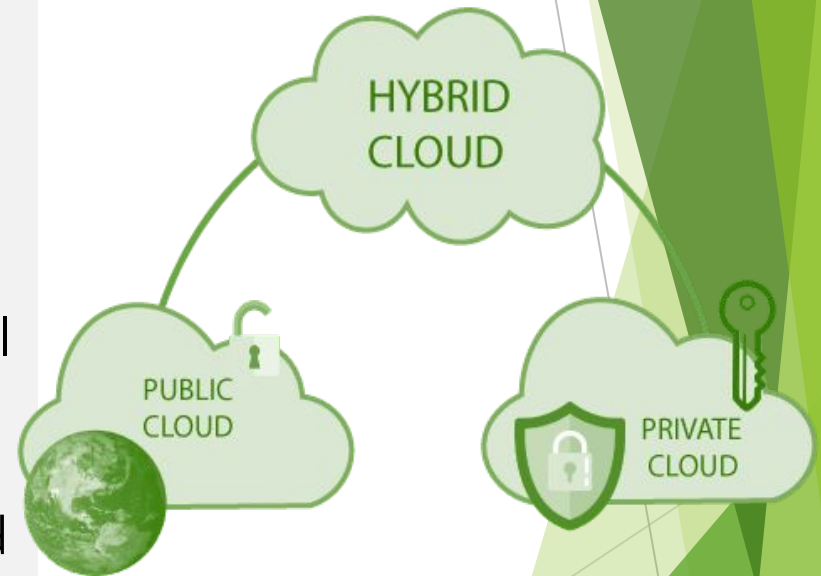
- Infrastructure is dedicated to a single organization and may be located on-premises or off-premises.
- Private cloud environments are designed to meet specific security, compliance, or performance requirements.
- They offer enhanced control, customization, and privacy but require significant upfront investment.



Cloud Deployment Models

Hybrid Cloud

- Combines public and private cloud environments, allowing for flexibility and data sharing between the two.
- Organizations can leverage the benefits of both public and private clouds, ensuring optimal resource allocation.
- Hybrid cloud deployments enable workload portability and seamless integration between different environments.



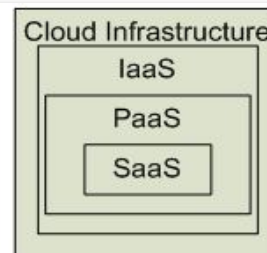
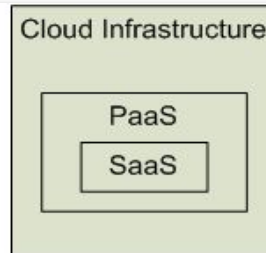
Software as a Service (SaaS)

Platform as a Service (PaaS)

Infrastructure as a Service (IaaS)

SalesForce CRM

LotusLive

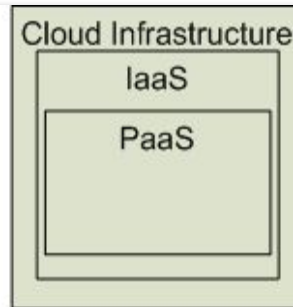
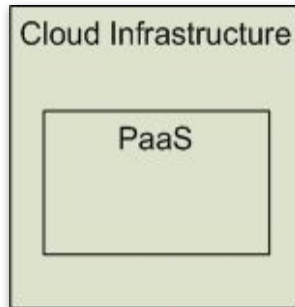


Software as a Service (SaaS)
Providers
Applications



Google App

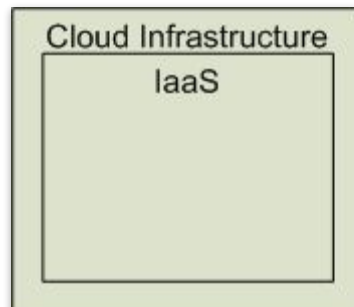
Windows Azure
The Future Made Familiar



Platform as a Service (PaaS)
Deploy customer
created Applications

amazon web services™

rackspace®
HOSTING



Infrastructure as a Service (IaaS)

Rent Processing, storage, N/W
capacity & computing resources

- Machines in the cloud
 - Can I move my VM elsewhere?
- Storage in the cloud
 - Can I move my data elsewhere?
- Databases in the cloud
 - Can I move my data elsewhere?
- Applications in the cloud
 - Can I run my application elsewhere?

There are four basic things people are doing in the cloud:

- Machines in the cloud
- Storage in the cloud
- Databases in the cloud
- Applications in the cloud

In addition to these four basics, cloud providers offer other services such as message queues and data mining. All of these things are lumped into the generic term “cloud computing”

Machines in the cloud

Many cloud providers allow you to create a Virtual Machine (VM) and deploy it in the cloud:

- Your VM images are stored in cloud storage
- You can create as many images as you need
- You can automatically start and stop running instances of those images as needed

Most cloud storage systems are designed as distributed, redundant systems

- Your data are stored on more than one disk in more than one place
- If one part of the system goes down, the rest of the system keeps going
- “There should never be a single point of failure” is a stated design goal

Cloud databases have similar design points

Datasets are distributed for reliability

- Some cloud databases support schemas, some don't
- Some cloud databases support joins, most don't
- Some cloud databases are relational, almost all aren't
- Some cloud databases are transactional, some aren't

- SaaS is a model of software deployment where an application is hosted as a service provided to customers across the Internet.
- SaaS settle-down the load of software maintenance/support
 - but users quit control over software versions and requirements.
- Terms that are used in this field include
 - **Platform as a Service (PaaS)** and
 - **Infrastructure as a Service (IaaS)**

- Cloud computing enables companies and applications, which are system infrastructure dependent, to be infrastructure-less.
- By using the Cloud infrastructure on “pay as used and on demand”, all of us can save in capital and operational investment!
- Clients can:
 - Put their data on the platform instead of on their own desktop PCs and/or on their own servers.
 - They can put their applications on the cloud and use the servers within the cloud to do processing and data manipulations



Amazon Elastic Compute Cloud (Amazon EC2) - Beta



TAP INTO THE
POWER OF NETWORK.COM



info@3tera.com (449) 305-0050

CAREERS | SU

APPLIC | UTILITY COMPUTING | TECHNOLOGY | PARTNERS | GRID UNIVERSITY | COMPANY

Cloud Computing

Cloudware - Cloud Computing Without Compromise



MOSSO
the hosting cloud



VERIO
An NTT Communications Company

- Several large Web companies are now exploiting the fact that they have data storage capacity that can be hired out to others.
 - allows data stored remotely to be temporarily cached on desktop computers, mobile phones or other Internet-linked devices.
- Amazon's Elastic Compute Cloud (EC2) and Simple Storage Solution (S3) are well known examples
 - Mechanical Turk

- The use of the cloud provides a number of opportunities:
 - It enables services to be used without any understanding of their infrastructure.
 - Cloud computing works using economies of scale:
 - It potentially lowers the outlay expense for start up companies, as they would no longer need to buy their own software or servers.
 - Cost would be by on-demand pricing.
 - Vendors and Service providers claim costs by establishing an ongoing revenue stream.
 - Data and services are stored remotely but accessible from “anywhere”.

- In parallel there has been backlash against cloud computing:
 - Use of cloud computing means dependence on others and that could possibly limit flexibility and innovation:
 - The others are likely become the bigger Internet companies like Google and IBM, who may monopolise the market.
 - Security could prove to be a big issue:
 - It is still unclear how safe out-sourced data is and when using these services ownership of data is not always clear.
 - There are also issues relating to policy and access:
 - If your data is stored abroad whose policy do you adhere to?
 - What happens if the remote server goes down?
 - How will you then access files?
 - There have been cases of users being locked out of accounts and losing access to data.

Advantages:

- Lower computer costs:
 - You do not need a high-powered and high-priced computer to run cloud computing's web-based applications.
 - Since applications run in the cloud, not on the desktop PC, your desktop PC does not need the processing power or hard disk space demanded by traditional desktop software.
 - When you are using web-based applications, your PC can be less expensive, with a smaller hard disk, less memory, more efficient processor...
 - In fact, your PC in this scenario does not even need a CD or DVD drive, as no software programs have to be loaded and no document files need to be saved.

Advantages:

- Improved performance:
 - With few large programs hogging your computer's memory, you will see better performance from your PC.
 - Computers in a cloud computing system boot and run faster because they have fewer programs and processes loaded into memory...
- Reduced software costs:
 - Instead of purchasing expensive software applications, you can get most of what you need for free-ish!
 - most cloud computing applications today, such as the Google Docs suite.
 - better than paying for similar commercial software which alone may be justification for switching to cloud applications

Advantages:

- Instant software updates:
 - Another advantage to cloud computing is that you are no longer faced with choosing between obsolete software and high upgrade costs.
 - When the application is web-based, updates happen automatically
 - available the next time you log into the cloud.
 - When you access a web-based application, you get the latest version
 - without needing to pay for or download an upgrade.
- Improved document format compatibility.
 - You do not have to worry about the documents you create on your machine being compatible with other users' applications or OSes
 - There are potentially no format incompatibilities when everyone is sharing documents and applications in the cloud.

Advantages:

- Unlimited storage capacity:
 - Cloud computing offers virtually limitless storage.
 - Your computer's current 1 Tbyte hard drive is small compared to the hundreds of Pbytes available in the cloud.
- Increased data reliability:
 - Unlike desktop computing, in which if a hard disk crashes and destroy all your valuable data, a computer crashing in the cloud should not affect the storage of your data.
 - if your personal computer crashes, all your data is still out there in the cloud, still accessible
 - In a world where few individual desktop PC users back up their data on a regular basis, cloud computing is a data-safe computing platform!

Advantages:

- Universal document access:
 - That is not a problem with cloud computing, because you do not take your documents with you.
 - Instead, they stay in the cloud, and you can access them whenever you have a computer and an Internet connection
 - Documents are instantly available from wherever you are
- Latest version availability:
 - When you edit a document at home, that edited version is what you see when you access the document at work.
 - The cloud always hosts the latest version of your documents
 - as long as you are connected, you are not in danger of having an outdated version

Advantages:

- Easier group collaboration:
 - Sharing documents leads directly to better collaboration.
 - Many users do this as it is an important advantages of cloud computing
 - multiple users can collaborate easily on documents and projects
- Device independence.
 - You are no longer tethered to a single computer or network.
 - Changes to computers, applications and documents follow you through the cloud.
 - Move to a portable device, and your applications and documents are still available.

Limitations:

- Requires a constant Internet connection:
 - Cloud computing is impossible if you cannot connect to the Internet.
 - Since you use the Internet to connect to both your applications and documents, if you do not have an Internet connection you cannot access anything, even your own documents.
 - A dead Internet connection means no work and in areas where Internet connections are few or inherently unreliable, this could be a deal-breaker.

Limitations:

- Does not work well with low-speed connections:
 - Similarly, a low-speed Internet connection, such as that found with dial-up services, makes cloud computing painful at best and often impossible.
 - Web-based applications require a lot of bandwidth to download, as do large documents.
- Features might be limited:
 - This situation is bound to change, but today many web-based applications simply are not as full-featured as their desktop-based applications.

Limitations:

- Can be slow:
 - Even with a fast connection, web-based applications can sometimes be slower than accessing a similar software program on your desktop PC.
 - Everything about the program, from the interface to the current document, has to be sent back and forth from your computer to the computers in the cloud.
 - If the cloud servers happen to be backed up at that moment, or if the Internet is having a slow day, you would not get the instantaneous access you might expect from desktop applications.

Limitations:

- Stored data might not be secure:
 - With cloud computing, all your data is stored on the cloud.
 - The questions is How secure is the cloud?
 - Can unauthorised users gain access to your confidential data?
- Stored data can be lost:
 - Theoretically, data stored in the cloud is safe, replicated across multiple machines.
 - But on the off chance that your data goes missing, you have no physical or local backup.
 - Put simply, relying on the cloud puts you at risk if the cloud lets you down.

Security Problems Associated with Cloud Computing

- ▶ Most security problems stem from:
 - ▶ Loss of control
 - ▶ Lack of trust (mechanisms)
 - ▶ Multi-tenancy
- ▶ These problems exist mainly in 3rd party management models
 - ▶ Self-managed clouds still have security issues, but not related to above

Loss of Control in the Cloud

- ▶ Consumer's loss of control
 - ▶ Data, applications, resources are located with provider
 - ▶ User identity management is handled by the cloud
 - ▶ User access control rules, security policies and enforcement are managed by the cloud provider
 - ▶ Consumer relies on provider to ensure
 - ▶ Data security and privacy
 - ▶ Resource availability
 - ▶ Monitoring and repairing of services/resources

Lack of Trust in the Cloud

- ▶ A brief deviation from the talk
 - ▶ Trusting a third party requires taking risks
- ▶ Defining trust and risk
 - ▶ Opposite sides of the same coin (J. Camp)
 - ▶ People only trust when it pays (Economist's view)
 - ▶ Need for trust arises only in risky situations

Multi-tenancy Issues in the Cloud

- ▶ Conflict between tenants' opposing goals
 - ▶ Tenants share a pool of resources and have opposing goals
- ▶ How does multi-tenancy deal with conflict of interest?
 - ▶ Can tenants get along together and 'play nicely' ?
 - ▶ If they can't, can we isolate them?
- ▶ How to provide separation between tenants?

Security Issues in the Cloud

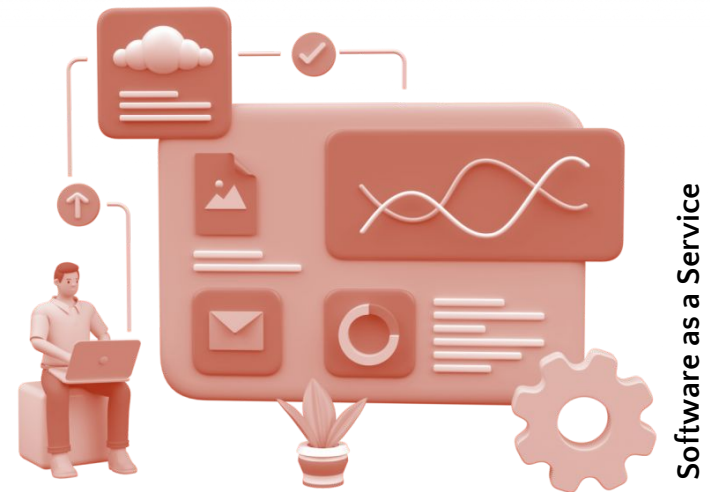
- ▶ In theory, minimizing any of the issues would help:
 - ▶ Loss of Control
 - ▶ Take back control
 - ▶ Data and apps may still need to be on the cloud
 - ▶ But can they be managed in some way by the consumer?
 - ▶ Lack of trust
 - ▶ Increase trust (mechanisms)
 - ▶ Technology
 - ▶ Policy, regulation
 - ▶ Contracts (incentives): topic of a future talk
 - ▶ Multi-tenancy
 - ▶ Private cloud
 - ▶ Takes away the reasons to use a cloud in the first place
 - ▶ VPC: its still not a separate system
 - ▶ Strong separation

What is SaaS?

SaaS stands for Software as a Service.

It is a cloud computing model where software applications are provided to users over the internet. Users access the software through a web browser, eliminating the need for installation and maintenance on their local machines.

With SaaS, users can remotely access the software without the need for installation on their local devices. This model allows businesses to subscribe to the software service with a monthly fee and a pay-as-you-go approach, providing flexibility and cost-effectiveness.

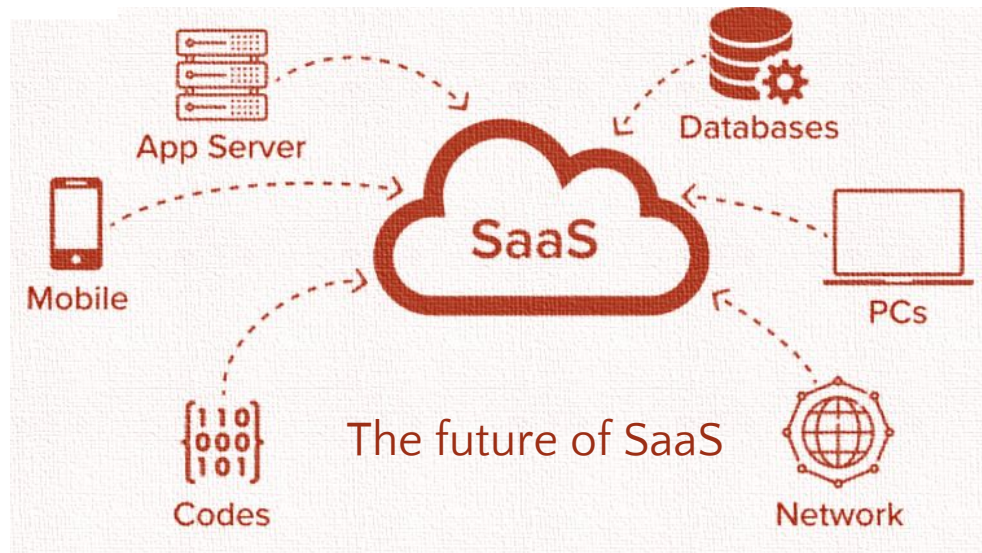


Software as a Service

Cont...

How Does SaaS Work?

SaaS providers host and manage the software applications on their servers. Users access the applications through a client interface, usually a web browser. The provider takes care of software updates, security, and infrastructure management, allowing users to focus on using the software rather than maintaining it.



Benefits of SaaS

Benefits of SaaS

- **Cost Savings:** SaaS eliminates the need for upfront hardware and software investments, reducing costs for businesses.
- **Scalability:** SaaS allows users to easily scale up or down based on their requirements, paying only for the resources they use.
- **Accessibility:** Users can access SaaS applications from any device with an internet connection, enabling remote work and collaboration.
- **Easy Updates:** SaaS providers handle software updates and patches, ensuring users always have access to the latest features and improvements.

Advantages of SaaS for Businesses

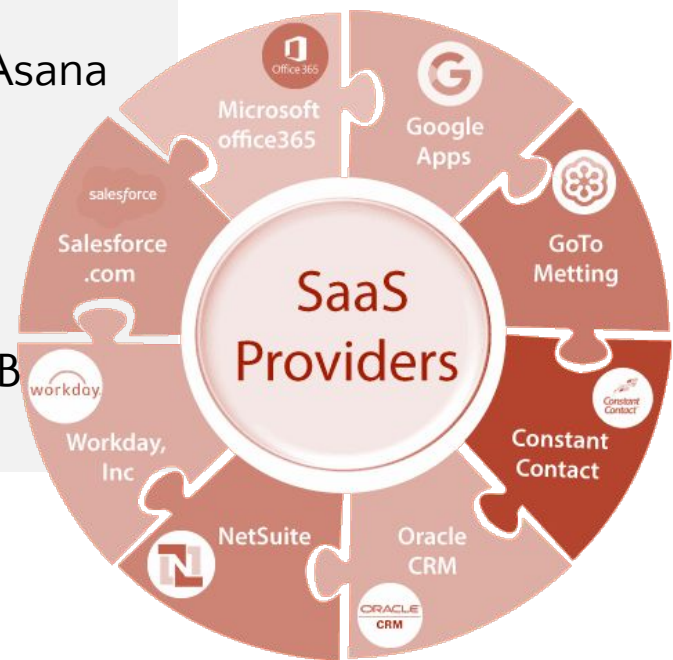
Advantages for Businesses

- **Rapid Deployment:** SaaS applications can be quickly deployed, reducing time-to-market for businesses.
- **Lower Maintenance:** With SaaS, businesses don't need to worry about software maintenance, updates, or infrastructure management.
- **Enhanced Collaboration:** SaaS applications often include collaboration features, enabling teams to work together seamlessly.
- **Improved Security:** SaaS providers implement robust security measures to protect user data, often surpassing what individual businesses can achieve.

SaaS Applications

Examples of SaaS Applications

- **Customer Relationship Management (CRM)** systems like Salesforce.
- **Project Management** tools such as Asana and Trello.
- **Communication and Collaboration** platforms like Slack and Google Workspace.
- **Accounting Software** such as QuickBooks Online and Xero.



SaaS vs Traditional Software

SaaS vs. Traditional Software

- **SaaS:** Pay-as-you-go pricing, no upfront costs, automatic updates, scalable, accessible from anywhere.
- **Traditional Software:** Upfront license fees, installation required, manual updates, limited scalability, tied to specific devices.

SaaS Adoption Considerations

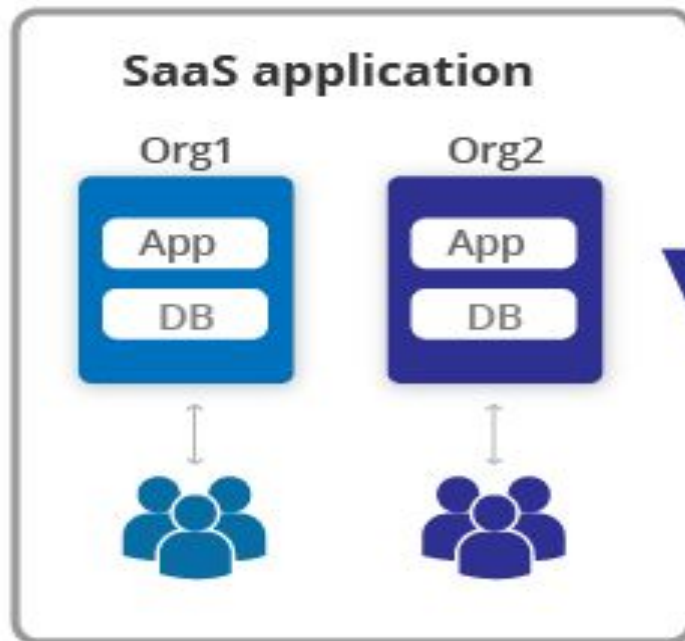
Data Security: Assess the security measures implemented by the SaaS provider to protect your sensitive data.

Service Level Agreements (SLAs): Review the SLAs to ensure the availability and performance of the SaaS application meet your business needs.

Integration Capabilities: Check if the SaaS application can integrate with your existing systems and workflows.

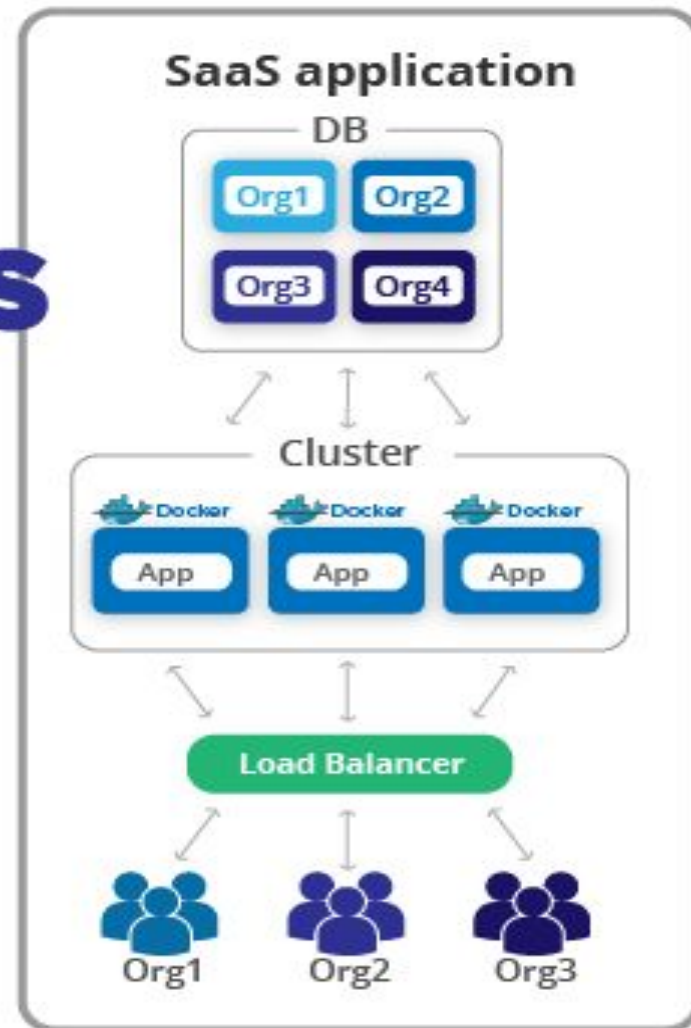
Vendor Reputation: Research the vendor's reputation, customer reviews, and track record before committing to a SaaS solution.

Single-Tenant



VS

Multi-Tenant



Multitenant Nature of SaaS

What is Multi tenant Architecture?

A single-tenant architecture (siloe model) is a single architecture per organization where the application has its own infrastructure, hardware, and software ecosystem.

Let's say you have ten organizations; in this case, you would need to create ten standalone environments, and your SaaS application or company will function as a single tenant architecture.

Multi tenant architecture is an ecosystem or model in which a single environment can serve multiple tenants utilizing a scalable, available, and resilient architecture. The underlying infrastructure is completely shared, logically isolated, and with fully centralized services.

Multi-tenant Architecture Benefits

a) A reduction of server Infrastructure costs utilizing a Multi tenant architecture strategy.

Instead of creating a SaaS environment per customer, you include one application environment for all your customers.

b) One single source of trust.

With a multi tenant SaaS architecture, you avoid this type of conflict, where you'll have one codebase (source of trust), and a code repository with a few branches (dev/test/prod). By following below practices, –with a single command (one-click-deployment)–, you will quickly perform the deployment process in a few seconds.

c) Cost reductions of development and time-to-market.

Cost reduction considers a sequence of decisions to make, such as having a single codebase, a SaaS platform environment, a multi-tenant database architecture, a centralized storage, APIs, and following will allow you to reduce development labor costs, time-to-market, and operational efficiencies.

"Multi-Tenant SaaS"

- "Multi-tenant SaaS is a business structure where many organizations share the same software to save and store data.
- Multi-tenant SaaS also implies that a single instance of the software and its supporting information is used by multiple customers.
- Each customer shares the same database and application.
- Consider the multi-tenant architecture as a high-rise building where the floor plans are prepared, but only a few changes can be made to individual units.
- It costs the customer a lot of money and time to make a significant change.

Benefits of SaaS Multi-Tenant Architecture

- **Lower costs through economies of scale:**

With multi-tenancy, scaling has far fewer infrastructure implications than with a single-tenancy-hosted.

- **Shared infrastructure leads to lower costs:**

- SaaS allows companies of all sizes to share infrastructure and data center operational costs.
- There is no need to add applications and more hardware to their environment.
- Not having to provision or manage any infrastructure or software above and beyond internal resources enables businesses to focus on everyday tasks.

- **Ongoing maintenance and updates:**
 - Customers don't need to pay costly maintenance fees to keep their software up to date.
 - Vendors roll out new features and updates. These are often included with a SaaS subscription.
- **Configuration can be done while leaving the underlying codebase unchanged:**
 - Single-tenant-hosted solutions are often customized, requiring changes to an application's code.
 - This customization can be costly and can make upgrades time-consuming because the upgrade might not be compatible with your environment.

- Multi-tenant solutions are designed to be highly configurable so that businesses can make the application perform the way they want.
- There is no changing the code or data structure, making the upgrade process easy.
- Multi-tenancy architecture also allows SaaS applications to efficiently service everyone from small customers, whose scale may not warrant dedicated infrastructure, to large enterprises that need access to the cloud's virtually

Understanding SOA (Service Oriented Approach)

- Service-oriented architecture (SOA) is a method of software development that uses software components called services to create business applications.
- Each service provides a business capability, and services. Developers use SOA to reuse and can also use for to communicate with each other across platforms and languages.
- Developers use SOA to reuse can also communicate with each other across platforms and languages. services in different systems or combine several independent services to perform complex tasks.

- For example, multiple business processes in an organization require the user authentication functionality. Instead of rewriting the authentication code for all business processes, you can create a single authentication service and reuse it for all applications.
- Similarly, almost all systems across a healthcare organization, such as patient management systems and electronic health record (EHR) systems, need to register patients.
- Such systems can call a single, common service to perform the patient registration task.

Benefits of service-oriented architecture

- **Faster time to market**
 - Developers reuse services across different business processes to save time and costs.
 - They can assemble applications much faster with SOA than by writing code and performing integrations from scratch.
- **Efficient maintenance**
 - It's easier to create, update, and debug small services than large code blocks in monolithic applications.
 - Modifying any service in SOA does not impact the overall functionality of the business process.
- **Greater adaptability**
 - SOA is more adaptable to advances in technology. You can modernize your applications efficiently and cost effectively.
 - For example, healthcare organizations can use the functionality of older electronic health record systems in newer cloud-based applications.

The basic principles of service-oriented architecture

- **Loose coupling**
 - Services in SOA should be loosely coupled, having as little dependency as possible on external resources such as data models or information systems.
 - They should also be stateless without retaining any information from past sessions or transactions.
 - This way, if you modify a service, it won't significantly impact the client applications and other services using the service.
- **Abstraction**
 - Clients or service users in SOA need not know the service's code logic or implementation details. To them, services should appear like a black box.
 - Clients get the required information about what the service does and how to use it through service contracts and other service description documents.
- **Granularity**
 - Services in SOA should have an appropriate size and scope, ideally packing one discrete business function per service.
 - Developers can then use multiple services to create a composite service for performing complex operations.

Components in Service-Oriented Architecture

There are four main components in service-oriented architecture (SOA).

- **Service**

- Services are the basic building blocks of SOA. They can be private—available only to internal users of an organization—or public—accessible over the internet to all.
- **Individually, each service has three main features.**
 - **Service implementation**
 - The service implementation is the code that builds the logic for performing the specific service function , such as user authentication or bill calculation.
 - **Service contract**
 - The service contract defines the nature of the service and its associated terms and conditions, such as the prerequisites for using the service, service cost, and quality of service provided.

- **Service provider**
 - The service provider creates, maintains, and provides one or more services that others can use. Organizations can create their own services or purchase them from third-party service vendors.
- **Service consumer**
 - The service consumer requests the service provider to run a specific service. It can be an entire system, application, or other service. The service contract specifies the rules that the service provider and consumer must follow when interacting with each other. Service providers and consumers can belong to different departments, organizations, and even industries.
- **Service registry**
 - A service registry, or service repository, is a network-accessible directory of available services. It stores service description documents from service providers. The description documents contain information about the service and how to communicate with it. Service consumers can easily discover the services they need by using the service registry.

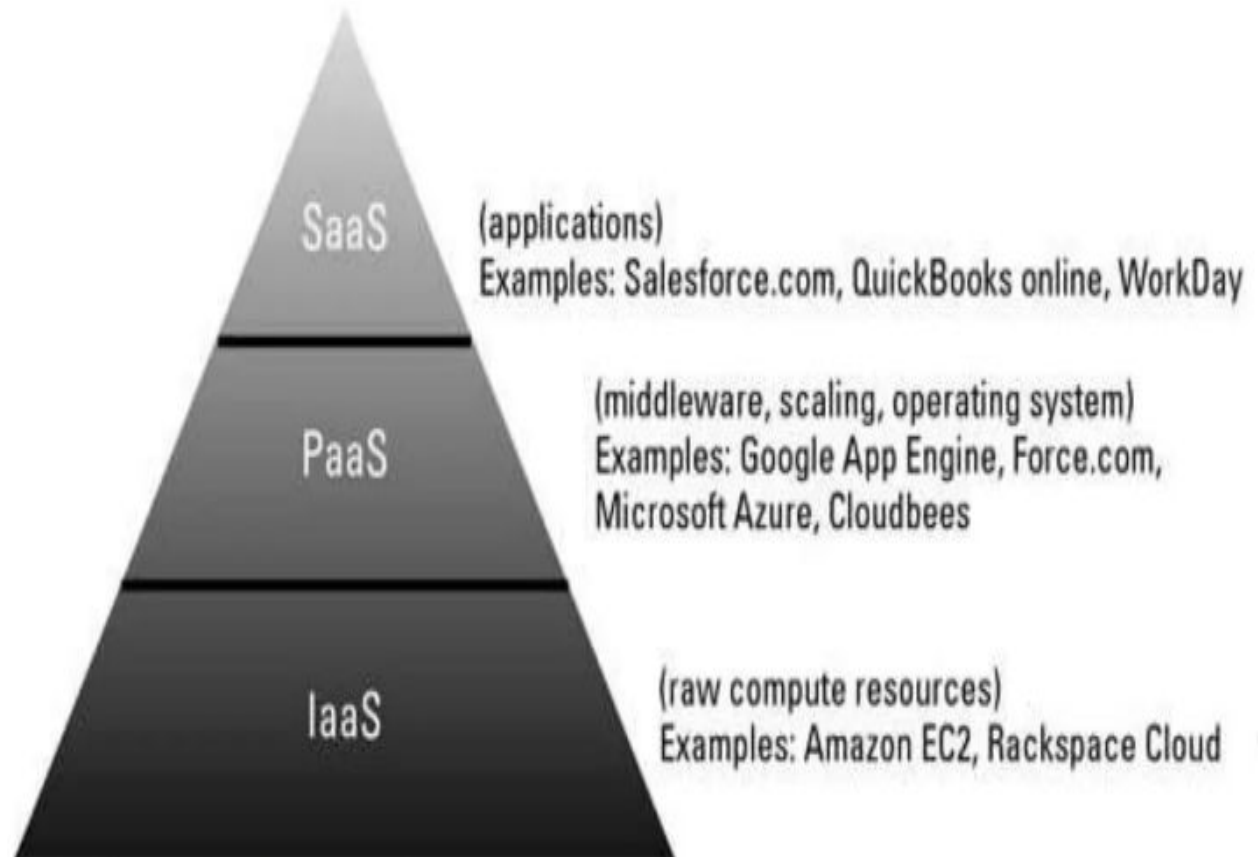
How does service-oriented architecture work?

- In service-oriented architecture (SOA), services function independently and provide functionality or data exchanges to their consumers.
- The consumer requests information and sends input data to the service. The service processes the data, performs the task, and sends back a response.
- For example, if an application uses an authorization service, it gives the service the username and password. The service verifies the username and password and returns an appropriate response.

Introduction to PaaS

PaaS in Cloud Computing

- Includes services for developing and deploying applications.
- It is an abstracted and integrated cloud-based computing environment that supports the development, running, and management of applications.
- A primary value of a PaaS environment is that developers don't have to be concerned with some of the lower-level details of the environment. You can look at a software stack as a pyramid:



The software stack as a pyramid.

What is PaaS ?

- PaaS, is a category of **cloud computing** that provides a **platform** and environment to allow developers to build applications and services over **the Internet**.
- A “cloud aware” **application development and deployment** environment.
- An **abstraction layer** between your cloud application and your IaaS provider.
- **PaaS services** are hosted in the **cloud** and accessed by users simply via their web browser.
- Fundamentally provides **elastic scaling of your application**.
- Deployments include **public, private and hybrid** cloud configurations.
- Example PaaS services:
 - Data services,
 - Application runtime,
 - Messaging & queueing,
 - Application management.

PaaS

- Platform as a Service (PaaS) is a way to rent hardware, operating systems, storage and network capacity over the Internet.
- The service delivery model allows the customer to rent virtualized servers and associated services for running existing applications or developing and testing new ones.
- It is an outgrowth of SaaS, a software distribution model in which hosted software applications are made available to customers over the Internet.

PaaS Goal

- Create an abstracted environment that supports an efficient, cost-effective, and repeatable process for the creation and deployment of high-quality applications.
- Focus on Development, not Ops
 - Programmers' development environment
 - Presentation layer: HTML, CSS, JavaScript
 - Control layer: Web Server code
 - Data layer: Data Model
 - Optionally, analytics
 - Ops below
 - Made visible through a web interface
 - Operating system
 - File system
 - User authentication
 - Utilities (cron, etc.)
 - Logs
 - Database maintenance, backups, etc.

How PaaS works

- Platform as a Service allows users to create software applications using tools supplied by the provider.
- PaaS services can consist of preconfigured features that customers can subscribe to; they can choose to include the features that meet their requirements while discarding those that do not.
- The infrastructure and applications are managed for customers and support is available.
- Services are constantly updated, with existing features upgraded and additional features added.

PaaS does that

- PaaS enables an organization to do the following:
- Leverage key middleware services without having to deal with the complexities of managing individual hardware and software elements.
- Access a complete stack of development and deployment tools via a web browser, a middleware environment where APIs can be used to plug into selected development and deployment tools. A developer might also leverage a full desktop development environment.
- Overcome the challenges of managing lots of individual development and deployment tools by providing a suite of integrated and standardized tools — operating systems, security products, and the like — that meet company requirements.

Payment

- As with most cloud offerings, PaaS services are generally paid for on a subscription basis with clients ultimately paying just for what they use.
- Clients also benefit from the economies of scale that arise from the sharing of the underlying physical infrastructure between users, and that results in lower costs.

PaaS offers

- Below are some of the features that can be included with a PaaS offering:
 - Operating system
 - Server-side scripting environment
 - Database management system
 - Server Software
 - Support
 - Storage
 - Network access
 - Tools for design and development
 - Hosting

Who can use PaaS services?

- Software developers, web developers and businesses can benefit from PaaS.
- For example, web developers can use individual PaaS environments at every stage of the process to develop, test and ultimately host their websites. However, businesses that are developing their own internal software can also utilise Platform as a Service, particularly to create distinct ring-fenced development and testing environments.

Primary Benefits of PaaS

- Lower development costs (%30+)
- Lower deployment and DevOps costs (Nearly eliminate)
- Application portability (depends on the PaaS)

Benefits for Developers

- Below are some of the benefits of PaaS to application developers:
 - They don't have to invest in physical infrastructure (being able to 'rent' virtual infrastructure)
 - They don't have to be concerned with some of the lower-level details of the environment.
 - Makes development possible for 'non-experts'
 - Flexibility
 - Adaptability
 - Teams in various locations can work together
 - Security

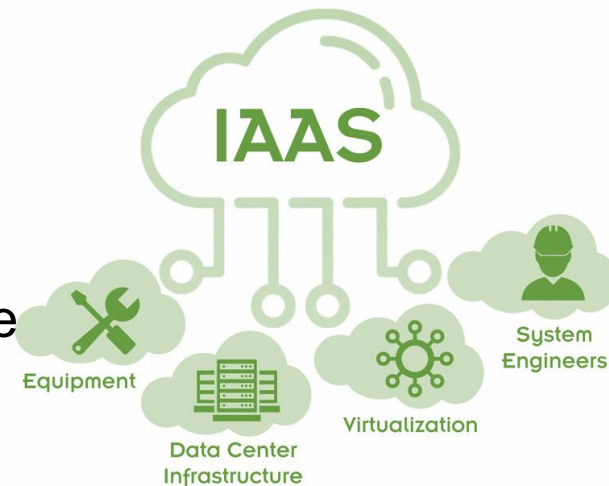
IaaS

What is IaaS?

Cloud infrastructure is the fundamental framework of hardware and software resources that empower cloud computing services. It encompasses essential components like servers, storage, networking, and virtualization enabling businesses to securely store, process and retrieve their data and applications in the cloud.

Cloud infrastructure refers to the foundation of hardware and software resources that enable cloud computing services.

It provides the necessary infrastructure components, such as servers, storage, networking, and virtualization, for businesses to store, process, and access data and applications in the cloud.



Key components of IaaS

IaaS comprises several essential components:

Virtual Machines (VMs): These are virtual instances of computers that provide computing power, including processing, memory, and storage, on demand.

Storage: IaaS offers scalable and reliable storage solutions that allow businesses to store and access data as needed.

Networking: IaaS provides networking capabilities, including virtual networks, load balancers, and firewalls, to connect and secure the infrastructure.

Management Tools: IaaS platforms offer management tools that allow users to provision, monitor, and manage their infrastructure efficiently.

Benefits of IaaS

IaaS offers numerous benefits for businesses:

- **Scalability:** IaaS enables businesses to scale their infrastructure up or down instantly, providing agility and cost optimization.
- **Cost Savings:** With IaaS, businesses can avoid the upfront costs of purchasing and maintaining physical hardware, paying only for the resources they consume.
- **Flexibility:** IaaS allows businesses to choose the infrastructure components they need and customize their environments to meet specific requirements.
- **Reliability:** IaaS providers ensure high availability and redundancy, minimizing downtime and providing reliable infrastructure services.

IaaS for Adoption

When adopting IaaS, What organizations should consider?

- **Security:** Implementing robust security measures to protect data and applications is essential. This includes access controls, encryption, and compliance with relevant regulations.
- **Performance:** Optimizing network performance, latency, and bandwidth is crucial to ensure a seamless user experience.
- **Vendor Selection:** Choose a reputable and reliable IaaS provider that aligns with your business needs and offers adequate support and service level agreements (SLAs).
- **Data Transfer and Integration:** Consider the ease of transferring data to and from the IaaS environment and integrating it with other systems or services.

Cloud Infrastructure Simplified with IaaS: Key Takeaways

- Cloud infrastructure serves as the foundation for cloud computing services, enabling businesses to store, process, and access data and applications in the cloud.
- IaaS provides virtualized computing resources, scalable storage, networking capabilities, and management tools.
- Key benefits of IaaS include scalability, cost savings, flexibility, and reliability.
- Security, performance, vendor selection, and data transfer are important considerations when adopting IaaS.